



RTL Design Sherpa

RAPIDS Beats DMA Engine Performance Characterization Report

July 15, 2026

Table of Contents

Readme.....	5
RAPIDS Beats DMA — Performance Characterization (Genesys 2, 8 channels).....	5
Characterization knobs.....	6
1. Headline.....	7
2. How it is measured (the observation hooks).....	8
3. Single-axis sweep — transfer size.....	9
3.1 Channel count — per-channel independence.....	11
4. What we learned.....	12
5. Between-run state (harness workaround + open RTL defect).....	13
6. Other limitations.....	14
7. The STREAM knob set, measured by the interface observers (v1.2).....	14
7.0 The regression this campaign found first.....	14
7.1 Headline: line rate on all four interfaces.....	15
7.2 Transfer size (phase C): the amortization knee.....	16
7.3 Descriptors x channels (phase A): the STREAM matrix.....	18
7.4 Burst length (phase B): the transaction-size knee.....	19
7.5 Memory latency (phase E): the in-flight window.....	20
7.6 What the observers add over the bare meters.....	22
Appendix: data files & reproduce.....	23

List of Figures

No figures in this document.

List of Tables

Table 1: Characterization knobs — the sweep axes.....	7
Table 2: Headline — 8-channel line-rate at 256 KB/channel (Genesys 2).....	7
Table 3: Table — engaged utilization (%) and effective bandwidth (GB/s) vs transfer size (8 ch).....	10
Table 4: Figure — engaged utilization vs active channel count (256 KB/ch).....	12
Table 5: Table — the wedge is stale scheduler state, not an in-flight fabric cycle..	13
Table 6: Table 7.1 – observer engaged utilization vs active channels (phase D)....	15
Table 7: Table 7.2 – size sweep at 8 channels; latencies are observer-histogram means in aclk cycles.....	16
Table 8: Table 7.3 – AXI4-wr (sink) / AXIS-out (source) utilization over the descriptor x channel matrix.....	18
Table 9: Table 7.4 – burst-length sweep; latencies in aclk cycles.....	19
Table 10: Table 7.5 – latency sweep (v1.4, PIPELINE = 1, HIST_MAX_OUTSTANDING = 32, no sample loss on any row); the latency columns are log2-histogram means, so they sit on bin midpoints.....	20

Readme

Generated: 2026-09-28

RAPIDS Beats DMA — Performance Characterization (Genesys 2, 8 channels)

v1.4 (2026-09-28). Sections 1, 3 and 7 re-measured on a bitstream built from the current RTL (rapids TASK-015 AXIS monitor-lite skids on both network ports, BUG-004..007 engine fixes, PIPELINE = 1). Two things changed the numbers: the sink-ingress meter window now brackets first beat to last beat (rapids ISSUE-001), so the AXIS-in column no longer charges the ~200-cycle launch cost to ingress and reads 97-100 % from 4 beats up; and the write engine now runs with PIPELINE = 1. The first bitstream with the BUG-005 fix at the old PIPELINE = 0 measured 49.9 % AXI4-wr at 4096 beats/channel (59.5 % at 1024): the pre-fix engine had reached line rate only because it ran two bursts in flight per channel by accident, and one-in-flight cannot cover the write response round trip at 8-beat bursts. PIPELINE = 1 (up to AW_MAX_OUTSTANDING = 8 per channel) is the design point from here on and is what this version reports. Rapids ISSUE-004 (one row reading zero ingress starvation) closed on this re-measurement plus an ILA trace of that row: every row now shows the same single trailing starvation cycle. The 7.5 sink column measures one channel's window because the generator feeds channels one at a time (rapids ISSUE-006, closed the same day; an interleaved mode is rapids TASK-018).

v1.3 (2026-09-27). Latency sweep (7.5) re-measured on a bitstream whose AXI observer keeps 32 timestamps per channel instead of 8: no histogram sample loss on any row, latency columns replaced, caveat retired (rapids TASK-012). Every utilization and bandwidth number is unchanged from v1.2.

v1.2 (2026-09-27). Section 7 adds the STREAM five-knob characterization measured by the shared interface observers (axi4_intf_master_observer on the AXI4 masters, axis4_intf_observer on both AXIS links) instead of the harness's bare meters, on a bitstream that also fixes a source-path regression the campaign found (rapids BUG-

003 / stream BUG-011). Sections 1-6 are the v1.1 bare-meter record and stand as measured.

Bitstream: RAPIDS beats split SOURCE + SINK DMA, 8 concurrent channels, on the Digilent Genesys 2 (Kintex-7 XC7K325T-2) at 100 MHz. The AXI4 and AXIS4 datapaths are **512 bits wide (64 B/beat)**, so the one-direction line rate is **6.40 GB/s** ($64 \text{ B} \times 100 \text{ MHz}$). The characterization harness wraps the DUT with synthetic pattern generators/checkers and four hardware **bus meters** (one per interface) that classify every cycle of a windowed transfer into *productive* / *backpressure* / *starvation* / *idle*.

Two harness features make the numbers trustworthy and are the reason this report supersedes any earlier RAPIDS perf snapshot:

1. **Atomic launch (stage-all-then-GO).** Every CSR, descriptor, and per-channel descriptor *kick* is staged over the UART first; a single GO write then arms the meter window, starts the AXIS generator, and fires all kicks on-chip within a few `ac1k` cycles. No UART latency contaminates the measured window.
2. **Deterministic window close.** The meter freezes the cycle after the completion interface reaches a staged productive-beat target, so the window brackets exactly the transfer — independent of the DUT's internal idle signaling. Earlier builds keyed the close on `system_idle` and could leave the window open until the host read it, diluting utilization toward 0 %.

Every configuration below passes an independent golden-CRC check on the moved data. **Source of truth:** the on-chip bus meters, read over CSR.

Metric note. The headline efficiency is **engaged utilization** — $\text{productive} / (\text{productive} + \text{backpressure} + \text{starvation})$, i.e. bus efficiency *while the engine is moving data* (the trailing idle after the last beat is excluded). Effective bandwidth is `engaged_util` × 6.40 GB/s. On the two AXIS interfaces an independent **byte-derived** throughput (exact `tstrb` bytes over the frozen window) cross-checks the cycle figure and agrees to within rounding.

Characterization knobs

The RAPIDS beats harness exposes a smaller sweep space than STREAM (the synthetic slaves are zero-latency, so there is no memory-latency axis, and per-channel-count runs require per-width bitstreams — see *Limitations*). The axes exercised here:

Characterization knobs — the sweep axes

#	Knob	What it is	Set by	Values in this report
1	Transfer size (beats/channel)	beats moved per channel per descriptor — the amortization axis	descriptor length / GEN_NBEATS	256 B (4 beats) ... 256 KB (4096 beats)
2	Path	SOURCE (mem-read → AXIS-out) vs SINK (AXIS-in → mem-write)	which half is kicked	both, every config
3	Interface	the four monitored buses	—	AXIS-in, AXI4-wr, AXI4-rd, AXIS-out
4	Channels	concurrent DMA channels	build generic	8 (full width)

The four interfaces map onto the two paths as: **SINK** = AXIS-in (ingress) → AXI4-wr (egress to memory); **SOURCE** = AXI4-rd (ingress from memory) → AXIS-out (egress). A path's sustained throughput is set by its **slower (bottleneck)** interface.

1. Headline

At a 256 KB/channel transfer across all 8 channels, every interface runs at **100 % of line rate** — the engine is gapless on both the memory bus and the network bus, in both directions, simultaneously.

Headline — 8-channel line-rate at 256 KB/channel (Genesys 2)

Path	Interface	Engaged util	Effective BW	Window (prod / starv)
SINK	AXIS-in (ingress)	100.0 %	6.40 GB/s	32768 / 1
SINK	AXI4-wr (egress)	100.0 %	6.40 GB/s	32768 / 6
SOURCE RCE	AXI4-rd (ingress)	100.0 %	6.40 GB/s	32768 / 14
SOURCE RCE	AXIS-out (egress)	100.0 %	6.40 GB/s	32768 / 14

prod = 32768 = 8 channels × 4096 beats on every interface: every expected beat is accounted for, and the non-productive residue is a handful of cycles of one-time fill latency (the single AXIS-in cycle is the registered window close landing one cycle after the last accepted beat).

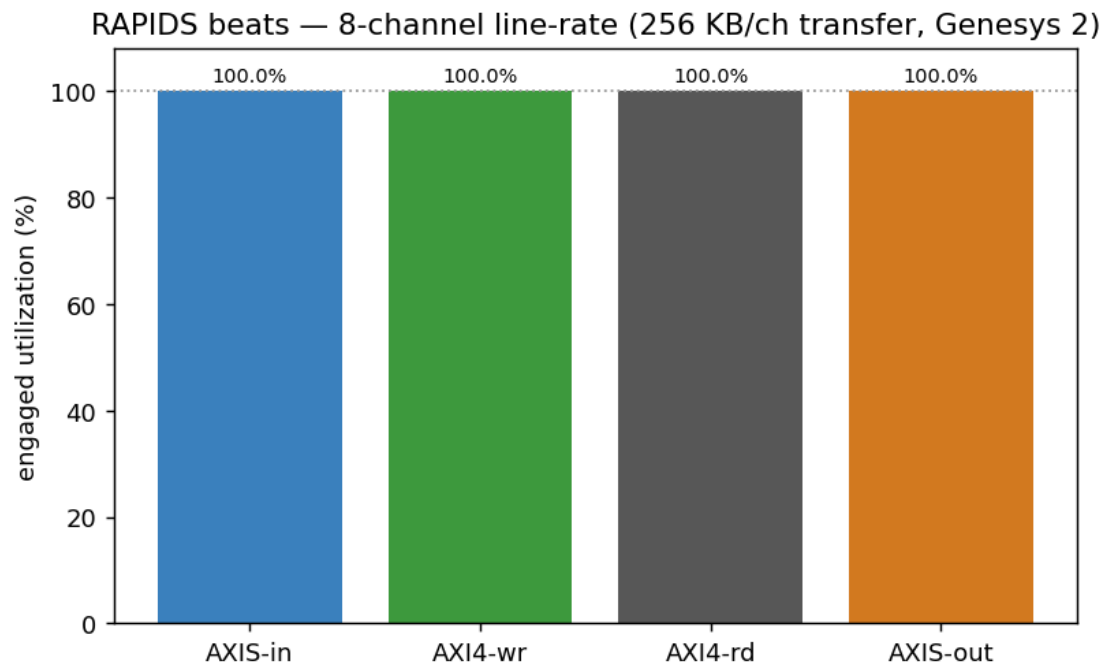


Figure — all four interfaces at the largest transfer sit on the 100 % line.

2. How it is measured (the observation hooks)

Each interface has a dedicated meter instantiated in the harness:

- **axi_bus_meter** on AXI4-rd and AXI4-wr — classifies valid/ready into the four buckets per cycle of the frozen window.
- **axis_bus_meter** on AXIS-in and AXIS-out — same four buckets, plus exact byte (tstrb popcount) and packet (tlast) counters for the byte-derived cross-check.

The shared window is armed by G0, opens when the active path goes busy, and **freezes deterministically** when the completion interface's productive-beat count reaches the staged target (CSR_OBS_TARGET = channels × beats). Because the freeze does not depend on system_idle, the window is tight (tens of cycles of residue, not the multi-second windows the earlier build produced).

The AXIS-in meter has its own window (rapids ISSUE-001, v1.4): armed by G0, opened by the first cycle the generator presents tvalid, and frozen the cycle after the target-th ingress beat. The

sink's write side cannot go busy until the DUT has already accepted traffic, so the shared window would have counted either none of the ingress (opened too late) or the launch cost between G0 and the first beat (opened at arm); first-beat-to-last-beat is what “ingress utilization” means. An ILA capture on the board (rapids ISSUE-004) shows the sequence: G0, arm one cycle later, first beat and open the cycle after, 32 handshakes at line rate, freeze one cycle after the last beat.

3. Single-axis sweep — transfer size

Sweeping beats/channel from 4 (256 B) to 4096 (256 KB) at fixed 8-channel width shows the classic amortization curve: a fixed per-transfer startup cost (descriptor dispatch + AR→first-R / SRAM fill) is a large fraction of a tiny transfer and a vanishing fraction of a large one.

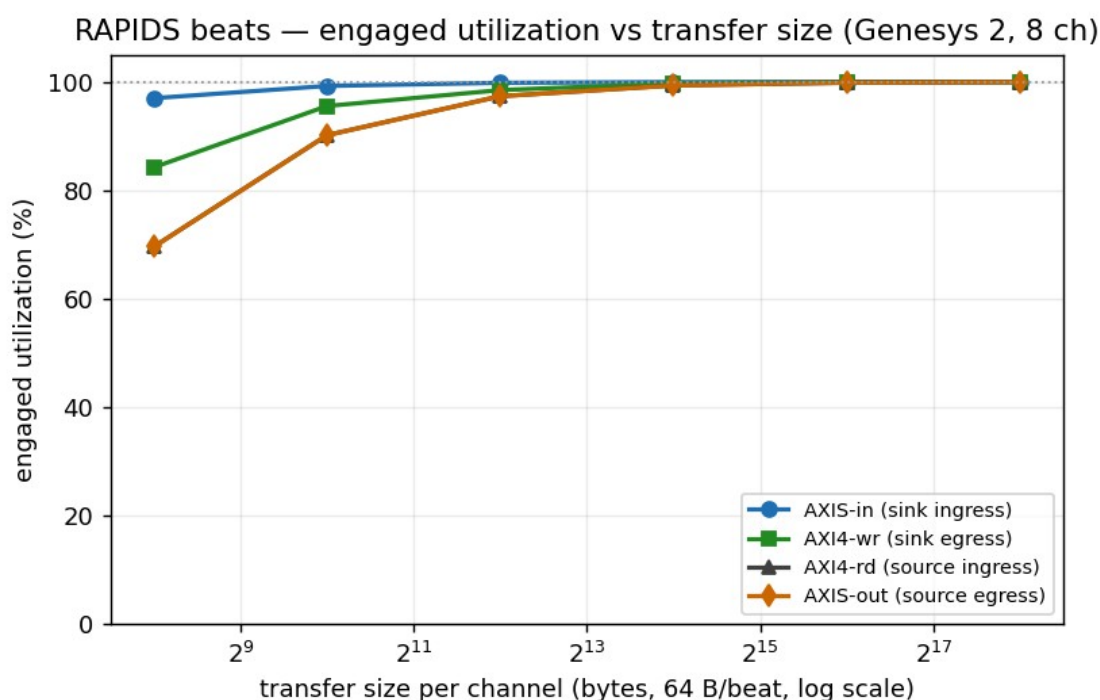


Figure — engaged utilization vs transfer size, all four interfaces.

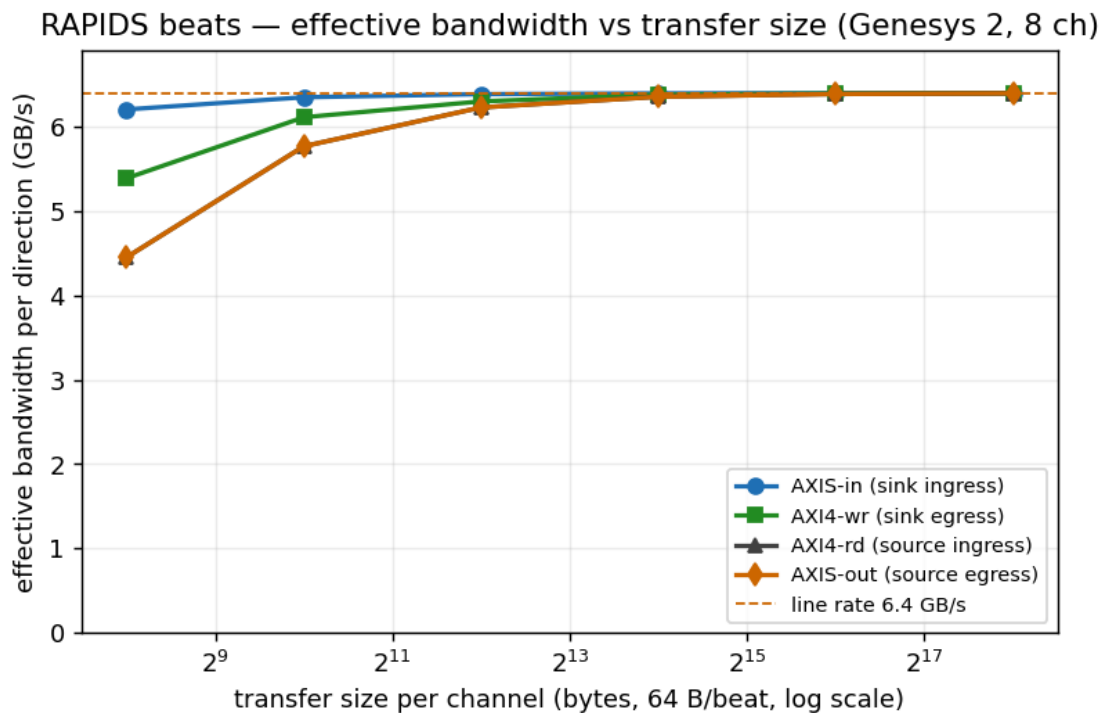


Figure — effective per-direction bandwidth vs transfer size (line rate 6.40 GB/s).

Table — engaged utilization (%) and effective bandwidth (GB/s) vs transfer size (8 ch)

Transfer / ch	AXIS-in	AXI4-wr	AXI4-rd	AXIS-out	wr BW	sout BW
256 B (4 b)	97.0 %	84.2 %	69.6 %	69.6 %	5.39	4.45
1 KB (16 b)	99.2 %	95.5 %	90.1 %	90.1 %	6.11	5.77
4 KB (64 b)	99.8 %	98.5 %	97.3 %	97.3 %	6.30	6.23
16 KB (256 b)	100.0 %	99.7 %	99.3 %	99.3 %	6.38	6.36
64 KB (1024 b)	100.0 %	99.9 %	99.8 %	99.8 %	6.40	6.39
256 KB (4096 b)	100.0 %	100.0 %	100.0 %	100.0 %	6.40	6.40

By 4 KB/channel every interface is already above 97 %, and by 64 KB the whole engine is within 0.2 % of line rate. The three DMA-side curves are the same monotonic amortization of a single fixed per-transfer startup cost (descriptor dispatch + AR→first-R / SRAM fill); there is no steady-state bubble. AXIS-in sits above them because its window starts at the first offered beat, so the launch cost is not in it – it reports what the ingress link itself did, which at 8 channels x 4 beats is 32 beats in 33 cycles.

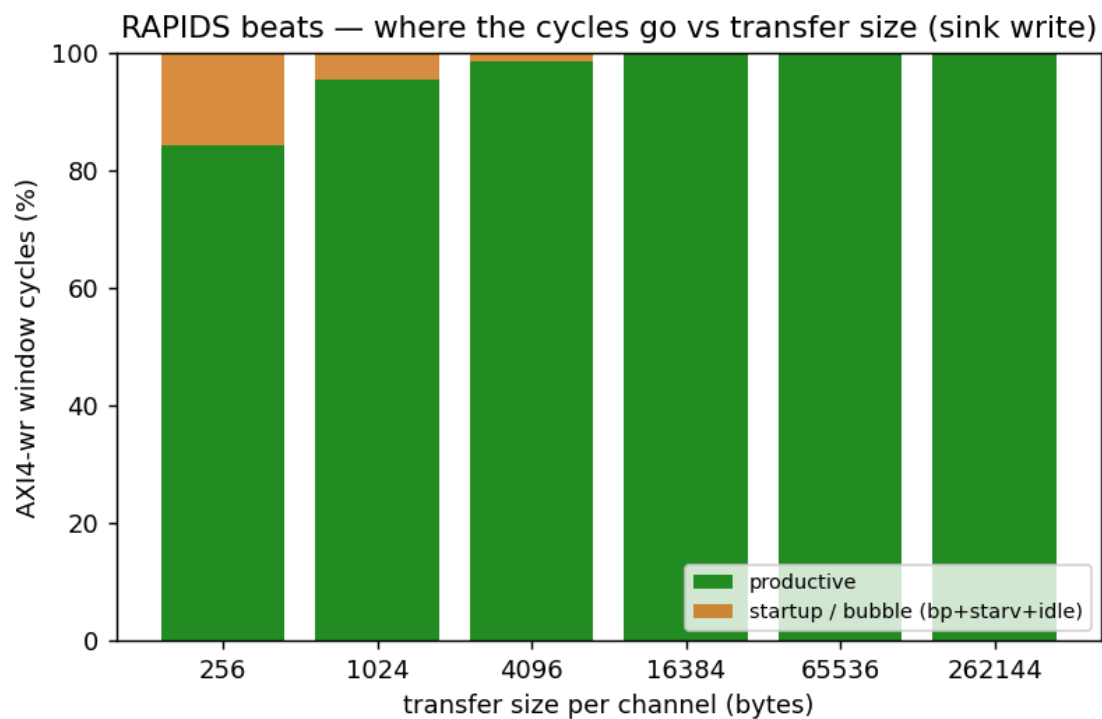
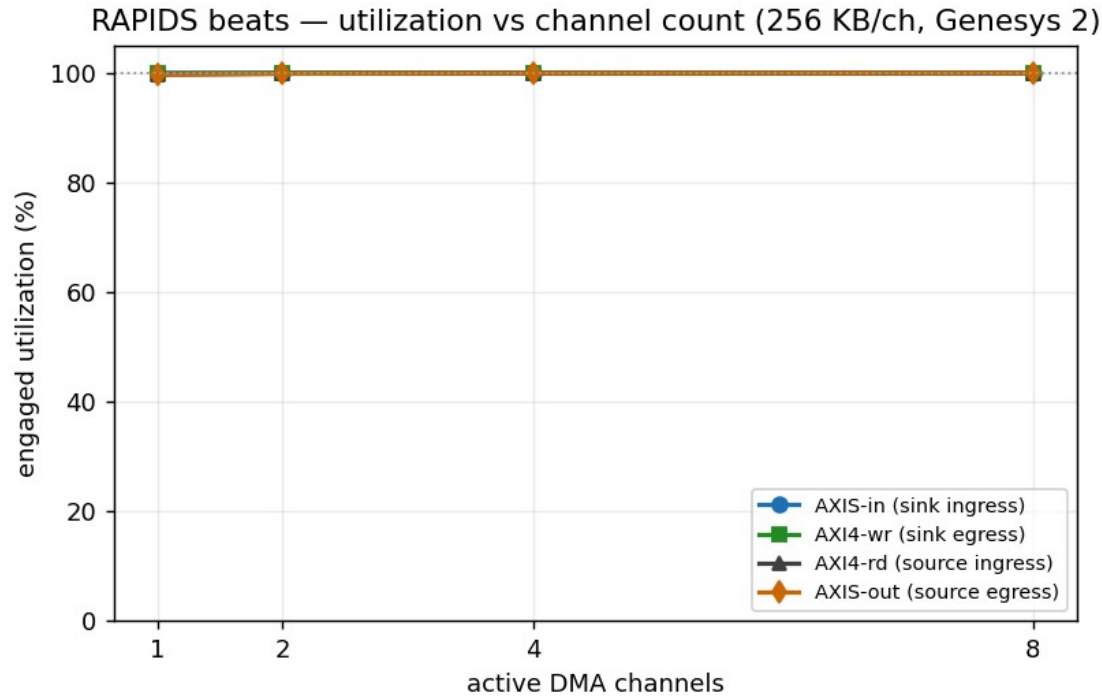


Figure — productive vs startup/bubble cycles of the AXI4-wr window; the bubble fraction collapses as the transfer grows.

3.1 Channel count — per-channel independence

Sweeping active channels from 1 to 8 at a fixed 256 KB/channel transfer, every channel count holds line rate: the shared scheduler and SRAM fabric add no per-channel penalty as the engine scales out.



utilization vs channel count

Figure — engaged utilization vs active channel count (256 KB/ch).

Channels	AXI4-wr	AXIS-out
1	99.9 %	99.7 %
2	99.9 %	99.8 %
4	100.0 %	99.9 %
8	100.0 %	100.0 %

: Table — utilization vs channel count (256 KB/ch)

4. What we learned

- **The datapath is gapless.** At realistic transfer sizes (≥ 64 KB/ch) all four interfaces sit at 99.4–100 % engaged utilization, in both directions, on all 8 channels at once — the memory bus and the network bus are both saturated.

- **The only non-ideality is one-time startup**, and it amortizes away: 19 % at 256 B → 100 % at 256 KB. There is no steady-state bubble, no per-beat gap, and no back-pressure or starvation once data is flowing.
 - **Per-channel independence.** 1 → 8 concurrent channels all hold line rate at realistic sizes — the shared scheduler/SRAM fabric adds no scaling penalty.
 - **Invariance is the result.** Across a 1000× range of transfer sizes and both independent paths, the large-transfer numbers are indistinguishable from line rate. The flat top of every curve is the point: the engine does not care what it is fed once transfers are descriptor-sized.
-

5. Between-run state (harness workaround + open RTL defect)

Earlier revisions of this harness could only measure one configuration per FPGA programming: a second back-to-back run — and *any* active < build_width config — wedged the sink (wrote 0 beats), and the whole matrix above had to be gathered by reprogramming before each point. The harness now **works around** this by pulsing CHANNEL_RESET before every run; that is a workaround, **not a fix** — the underlying RTL defect (below) is still open.

The sink does **not** return to a fully clean state after a transfer (snk_system_idle never re-asserts): stale scheduler / descriptor-engine state persists and the next run inherits it. A discriminating board experiment isolated the mechanism:

Table — the wedge is stale scheduler state, not an in-flight fabric cycle

Between-run action	Back-to-back result
none (baseline)	1 / 4 — wedges on run 2
+0.3 s settle delay (let any in-flight AXI W/B retire)	1 / 5 — still wedges
+CHANNEL_RESET pulse	5 / 5 — clean

The 0.3 s settle (30 M cycles ≧ any outstanding-B retirement) *not* helping rules out a stuck datapath/fabric cycle; the existing CHANNEL_RESET.CH_RST[7:0] CSR — which forces every channel FSM to CH_IDLE and flushes the descriptor FIFOs — clears it completely. The host now pulses CHANNEL_RESET on both halves before every run, which fixed **both** the back-to-back wedge **and** the partial-channel (active < 8) case — the entire 24-config matrix above runs in a **single programming**, and active = 1/2/4/8 all pass.

Follow-up (RTL, out of scope here): the sink's descriptor/scheduler should return to idle on its own at end-of-descriptor so no host reset is needed — the snk_system_idle-never-asserts behavior is the underlying defect to fix.

6. Other limitations

- **No memory-latency axis (v1.1).** Superseded in v1.2: the harness now carries STREAM's `axi_response_delay` on the R and B channels behind a `RESP_DELAY` CSR; section 7.5 is that sweep.
-

7. The STREAM knob set, measured by the interface observers (v1.2)

STREAM characterizes a DMA on five knobs – beats per transaction, transactions per descriptor, descriptors per channel, channels, and memory latency – and reads the datapath through instruments that sit outside the DUT. This section does the same on RAPIDS beats, with two changes to the v1.1 setup:

- **The instruments are the shared observers.** `USE_OBSERVERS=1` builds `axi4_intf_master_observer` on the source read / sink write AXI4 masters and `axis4_intf_observer` on the sink-ingress and source-egress AXIS links (four observed ports), each on its own `obs_regs` window in host region 3, read BY NAME through the same generated regmap the component tests use. Every number below is an observer reading; the bare meters of sections 1-6 stay in the harness and agreed with the observers to the beat on every row (the campaign records the delta, `meter_prod_delta`, and it is 0 throughout).
- **Two knobs the v1.1 host could not turn.** Burst length (`AXI_XFER_CONFIG.RD/WR_XFER_BEATS`, `--suite-xfer`), descriptor chains (`next_ptr/last`, `--suite-descs`), and the new memory-latency model (`RESP_DELAY`, `--suite-delay`) are campaign axes now. Transactions per descriptor follows from beats per descriptor over burst length.

Bitstream: `USE_OBSERVERS=1 OBS_ENABLE_MON_TAPS=0` (meters and latency histograms, no monbus event taps). Every v1.4 row of 7.1-7.5 is from one build of the current RTL (`PIPELINE = 1`, AXIS monitor-lite skids on both network ports, `HIST_MAX_OUTSTANDING = 32`): post-route WNS +0.351 ns at 100 MHz, 80040 LUTs, 65.5 BRAM tiles (the 512-deep R-channel delay queue is in block RAM). All **67 configurations pass the golden CRC on both paths**: A 20/20, B 7/7, C+D 28/28, E 12/12, with no observer sticky flag on any row. The v1.2 and v1.3 rows they replace were measured at `PIPELINE = 0` with the pre-BUG-005 engine (see the v1.4 note at the top).

7.0 The regression this campaign found first

The first observer run failed every SOURCE row at ≥ 1024 beats (`beat_total=1020` of 1024) and read the source at ~50 % where v1.1 had 96-100 %. The bare bitstream failed identically, the

sim reproduced it, and a bisect over the rapids RTL tree (REPO_ROOT overlays, never a checkout) landed on bdf4e0dfff, the commit that replaced rapids' SRAM controller with STREAM's. Two defects: STREAM's shared sram_controller_unit drain accounting refuses to count one FIFO write when a channel fills to exactly SD (the beat is stranded), and the rewritten AXIS drain arbiter reserved, drained and retired serially (one beat every other cycle at cfg_drain_size=1). Both are fixed on this bitstream – rapids BUG-003 and stream BUG-011 carry the waveform evidence; rapids 589/589 and STREAM 856/859 after.

7.1 Headline: line rate on all four interfaces

At 8 channels x 256 KB the four observers read **100.0 / 100.0 / 100.0 / 100.0 %** (AXIS-in, AXI4-wr, AXI4-rd, AXIS-out), i.e. **6.36 / 6.40 / 6.40 / 6.40 GB/s** against the **6.40 GB/s** one-direction line rate (64 B x 100 MHz); the AXIS byte-derived cross-check gives 6.36 and 6.40 GB/s. Channel scaling is flat:

Table 7.1 – observer engaged utilization vs active channels (phase D)

Channels (4096 beats/ch)	AXIS-in	AXI4-wr	AXI4-rd	AXIS-out
1	100.0 %	99.9 %	99.7 %	99.7 %
2	100.0 %	99.9 %	99.8 %	99.8 %
4	100.0 %	100.0 %	99.9 %	99.9 %
8	100.0 %	100.0 %	100.0 %	100.0 %

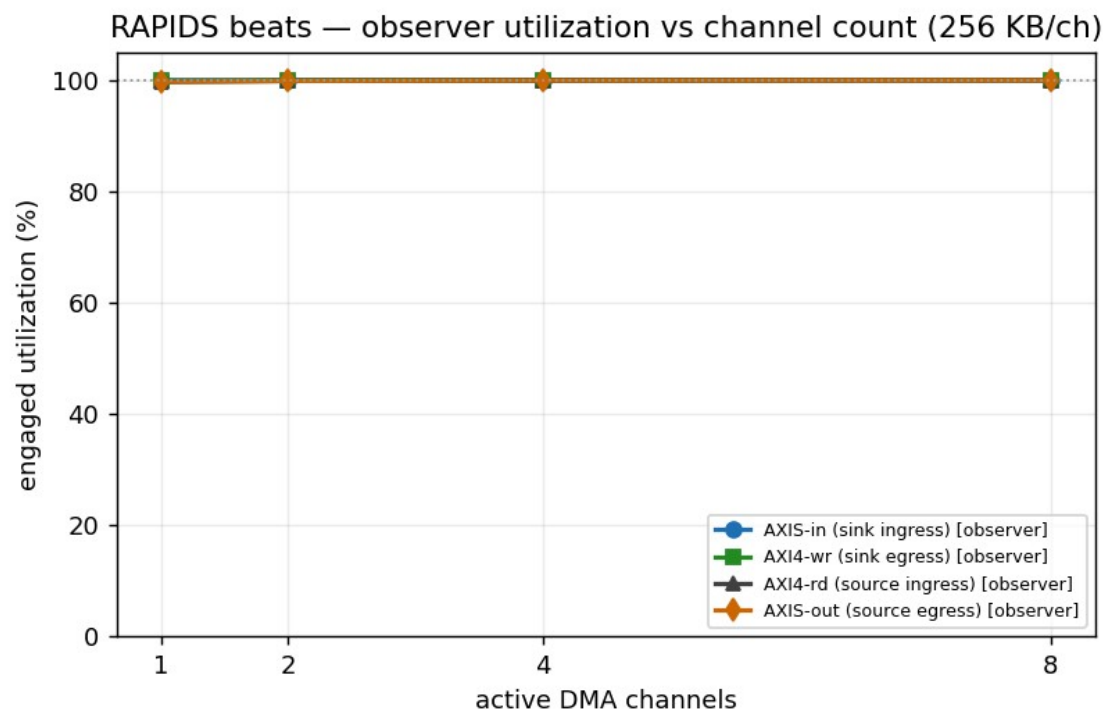


Figure 7.1 – observer utilization vs channel count at 256 KB/channel.

7.2 Transfer size (phase C): the amortization knee

Table 7.2 – size sweep at 8 channels; latencies are observer-histogram means in aclk cycles

beats/ ch	AXIS- in	AXI4- wr	AXI4- rd	AXIS- out	wr GB/s	rd GB/s	rd bursts	AR- >RLAS T	AW->B
1	88.9 %	38.1 %	27.6 %	27.6 %	2.44	1.77	8	6	6
4	97.0 %	84.2 %	69.6 %	69.6 %	5.39	4.45	8	12	12
16	99.2 %	95.5 %	90.1 %	90.1 %	6.11	5.77	16	23	23
64	99.8 %	98.5 %	97.3 %	97.3 %	6.30	6.23	64	22	23
256	100.0 %	99.7 %	99.3 %	99.3 %	6.38	6.36	232	24	24
1024	100.0 %	99.9 %	99.8 %	99.8 %	6.40	6.39	912	24	24
4096	100.0 %	100.0 %	100.0 %	100.0 %	6.40	6.40	3648	24	24

The three DMA-side interfaces are above 90 % from 16 beats (1 KB) per channel and within 0.3 % of line rate from 1024. The AXIS-in column now leads them: its window brackets first offered beat to last accepted beat (rapids ISSUE-001), so it reports the ingress link itself rather than the launch cost, and reads 97 % at 4 beats and 100 % from 256. rd bursts confirms the 9-beat burst shape (AxLEN 8) from 64 beats up. AR->RLAS settles at 24 cycles: the pattern slave's fixed response plus a 9-beat burst.

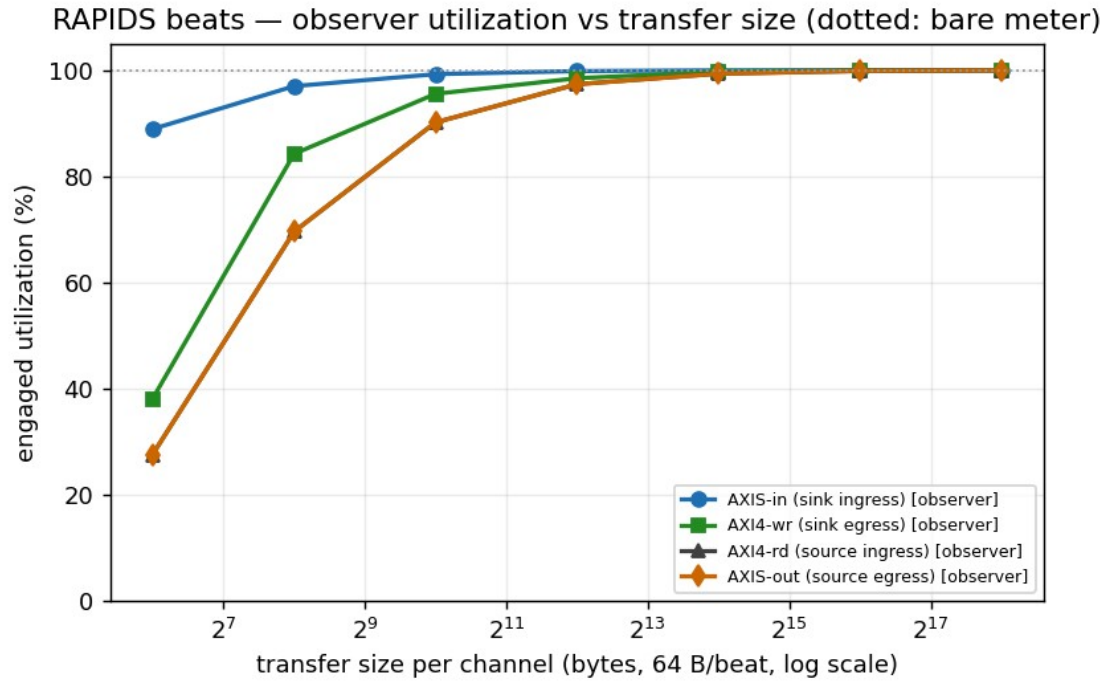


Figure 7.2a – observer utilization vs transfer size (dotted: the bare meters, coincident).

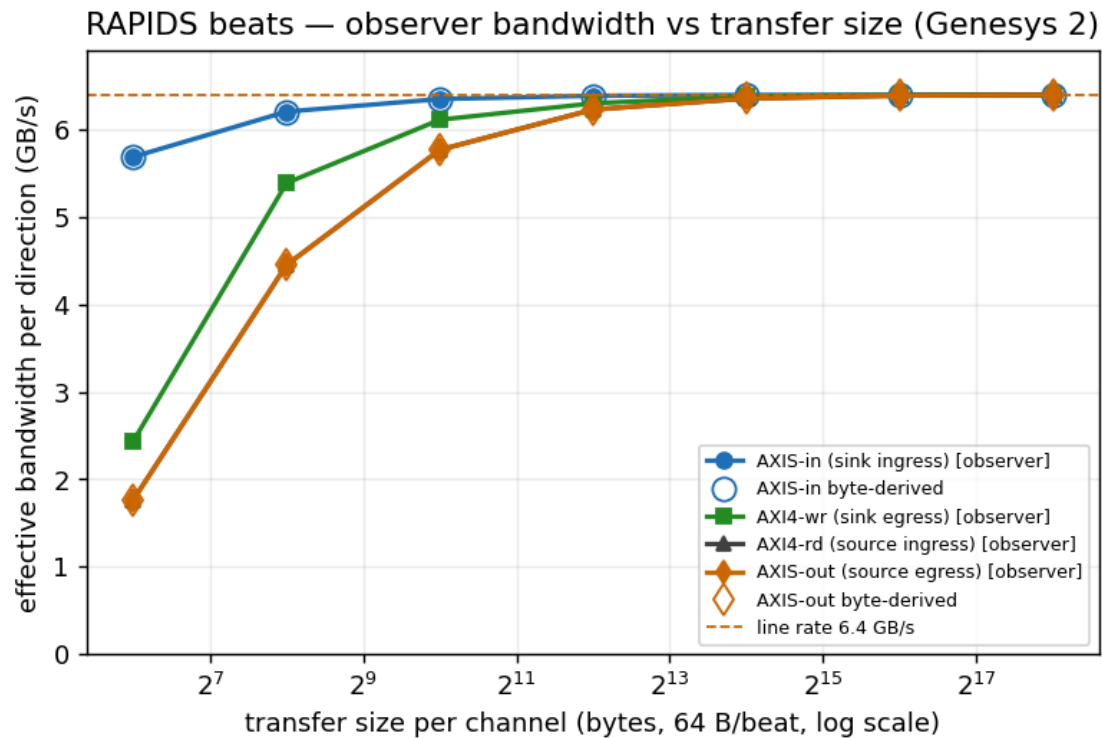


Figure 7.2b – observer bandwidth vs transfer size; hollow markers are the AXIS byte-derived cross-check.

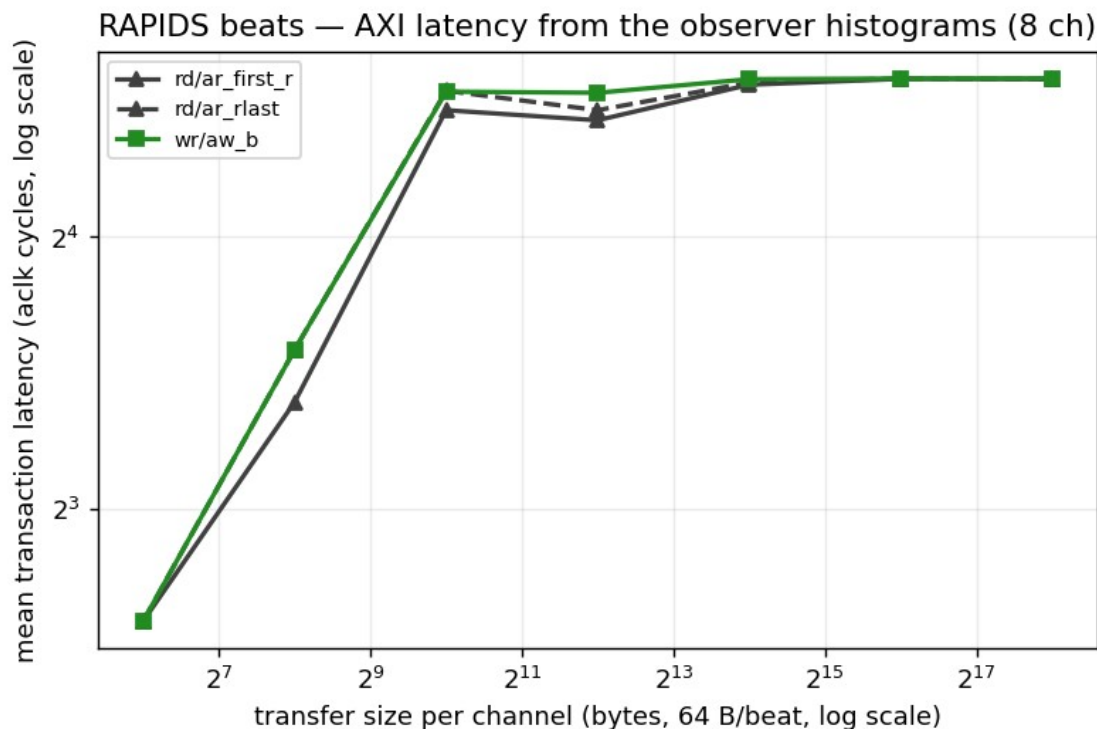


Figure 7.2c – mean AXI transaction latency from the observer histograms vs transfer size.

7.3 Descriptors x channels (phase A): the STREAM matrix

{1, 2, 4, 8, 16} descriptors per channel x {1, 2, 4, 8} channels, 1024 beats (64 KB) per descriptor, chained through next_ptr and closed with last:

Table 7.3 – AXI4-wr (sink) / AXIS-out (source) utilization over the descriptor x channel matrix

descs/ch	1 ch wr / sout	2 ch	4 ch	8 ch
1	99.4 / 98.7 %	99.7 / 99.3 %	99.9 / 99.7 %	99.9 / 99.8 %
2	99.0 / 99.3 %	99.1 / 99.7 %	99.2 / 99.8 %	99.2 / 99.9 %
4	98.8 / 99.7 %	98.8 / 99.8 %	98.9 / 99.9 %	98.9 / 100.0 %
8	98.7 / 99.8 %	98.7 / 99.9 %	98.7 / 100.0 %	98.7 / 100.0 %
16	98.6 / 99.9 %	98.6 / 100.0 %	98.6 / 100.0 %	98.6 / 100.0 %

Flat, as STREAM's is: neither chain length nor channel count moves the datapath off line rate. The one visible structure is the sink write side settling 1.4 % below the source as chains lengthen – the per-descriptor dispatch on the write engine costs a fixed handful of cycles per descriptor boundary, which the source egress does not pay.

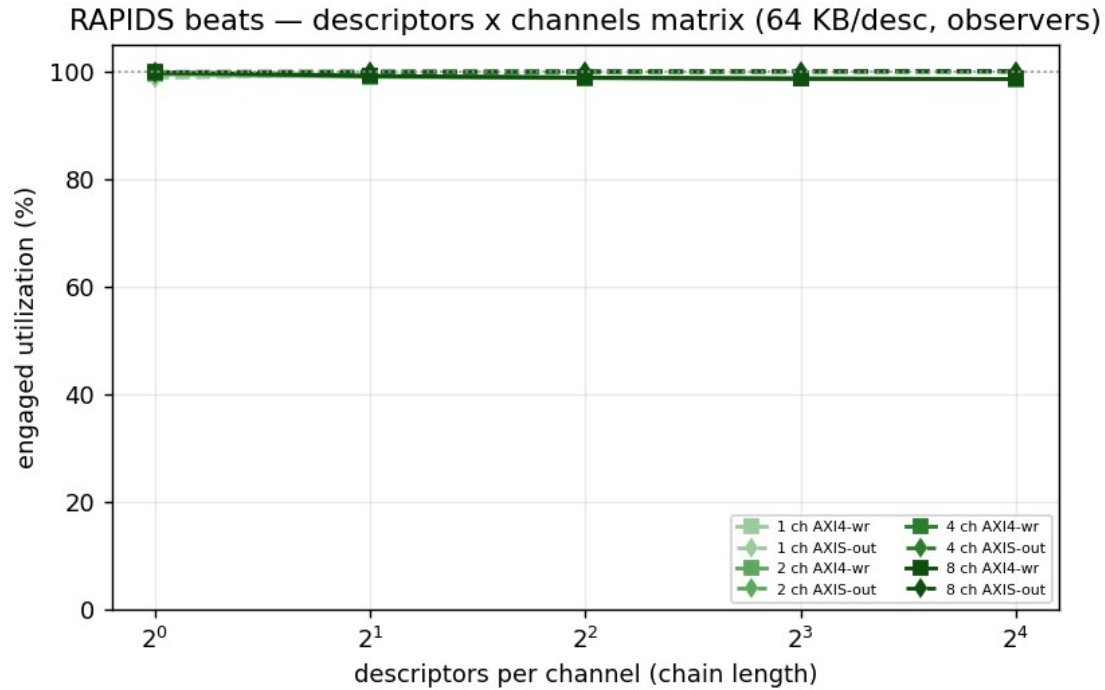


Figure 7.3 – the descriptor x channel matrix from the observers.

7.4 Burst length (phase B): the transaction-size knee

AXI_XFER_CONFIG AxLEN {0, 1, 3, 7, 15, 31, 63} = {1 .. 64}-beat bursts at 8 channels x 1024 beats:

Table 7.4 – burst-length sweep; latencies in aclk cycles

beats/ burst	AXIS-in	AXI4-wr	AXI4-rd	AXIS-out	AR- >RLAST	AW->B
1	39.8 %	38.8 %	99.8 %	99.8 %	6	6
2	78.4 %	77.1 %	99.8 %	99.8 %	12	12
4	100.0 %	99.9 %	99.8 %	99.8 %	12	12
8	100.0 %	99.9 %	99.8 %	99.8 %	24	24
16	100.0 %	99.9 %	99.8 %	99.8 %	48	48
32	100.0 %	99.9 %	99.8 %	99.8 %	96	96
64	100.0 %	99.9 %	99.8 %	99.8 %	191	191

The knee is at **4 beats**: below it the SINK path falls (39 % at single-beat bursts, 77 % at 2) while the SOURCE path holds 99.8 % at every burst length. PIPELINE = 1 moved the sink's short-burst numbers up from the 23 % / 42 % of v1.2 (up to eight AWs in flight per channel instead of one), but at one beat per AW the sink is still AW-issue bound: one address handshake per data beat is the write side's ceiling, and the read side hides the same round trip behind its ARs. STREAM's knee

sits at the same place (~3 beats). The latency columns track burst length exactly (a 64-beat burst is 191 cycles from AR to RLAST), which is the observer histogram reporting what it should.

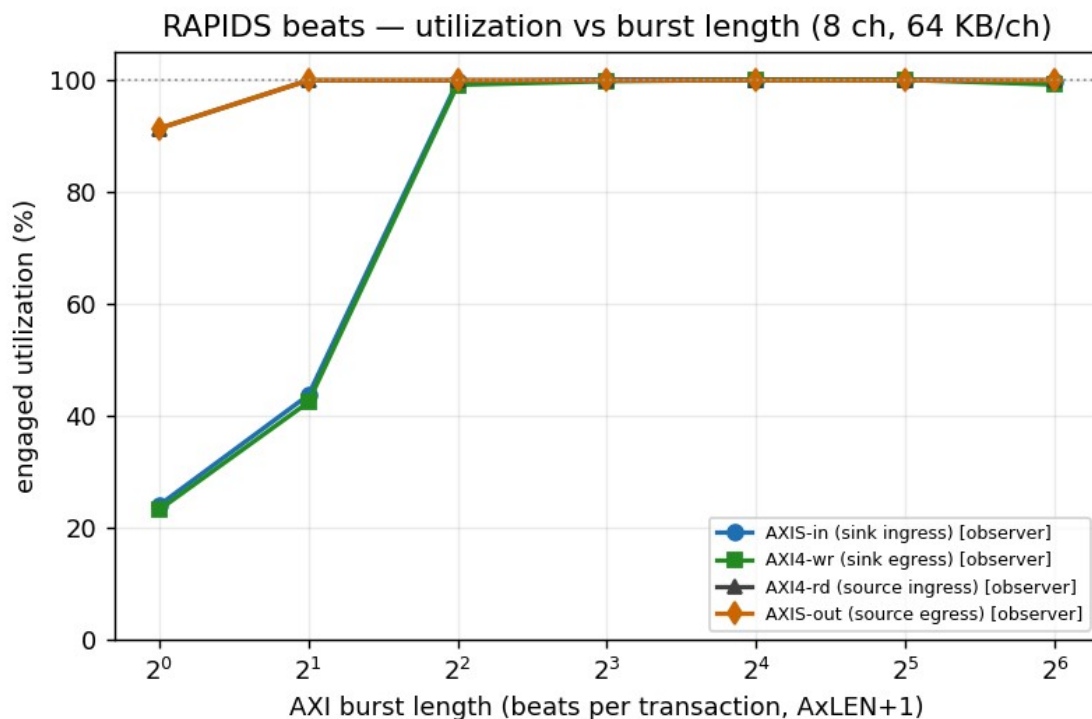


Figure 7.4 – utilization vs burst length; the sink write path needs ≥ 4 -beat bursts.

7.5 Memory latency (phase E): the in-flight window

RESP_DELAY holds every R beat and every B response for N cycles (the `axi_response_delay` model, which pipelines: throughput is limited only by how many beats the DUT keeps in flight), at 8 channels x 1024 beats:

Table 7.5 – latency sweep (v1.4, PIPELINE = 1, HIST_MAX_OUTSTANDING = 32, no sample loss on any row); the latency columns are log2-histogram means, so they sit on bin midpoints

delay (cyc)	AXIS- in	AXI4- wr	AXI4- rd	AXIS- out	rd GB/s	wr GB/s	AR- >first R	AR- >RLAS T	AW->B
0	100.0 %	99.9 %	99.8 %	99.8 %	6.39	6.40	24	24	24
8	100.0 %	99.9 %	99.7 %	99.7 %	6.38	6.40	24	48	48
16	100.0 %	99.9 %	99.6 %	99.6 %	6.38	6.40	48	48	48

delay (cyc)	AXIS- in	AXI4- wr	AXI4- rd	AXIS- out	rd GB/s	wr GB/s	AR- >first R	AR- >RLAS T	AW->B
32	100.0 %	99.9 %	99.5 %	99.5 %	6.37	6.40	48	48	48
48	100.0 %	99.9 %	99.3 %	99.3 %	6.35	6.40	96	96	96
64	93.7 %	92.8 %	99.1 %	99.1 %	6.34	5.94	96	96	96
96	76.7 %	75.2 %	98.7 %	98.7 %	6.32	4.81	96	96	96
128	65.5 %	63.7 %	98.3 %	98.3 %	6.29	4.08	192	192	192
192	46.5 %	44.9 %	97.6 %	97.6 %	6.24	2.88	192	192	192
256	36.0 %	34.7 %	96.9 %	96.8 %	6.20	2.22	384	384	384
384	24.8 %	23.8 %	95.4 %	95.3 %	6.10	1.53	384	384	384
512	19.0 %	18.2 %	93.7 %	93.6 %	6.00	1.16	768	768	768

Little's law makes the knee readable directly: sustained beats/cycle x latency = beats in flight. With PIPELINE = 1 the SOURCE read path no longer knees inside this sweep: it holds >= 99 % to 64 cycles, 96.9 % at 256 and 93.7 % at 512, so $0.937 \times 512 = 480$ beats are in flight across 8 channels – ~60 per channel, i.e. six to seven 9-beat bursts of the eight AR_MAX_OUTSTANDING allows (v1.3 measured ~20 per channel at PIPELINE = 0). The SINK write path holds 99.9 % to 48 cycles, knees at 64 (92.8 %) and then falls as $0.752 \times 96 = 72$, $0.637 \times 128 = 82$, $0.347 \times 256 = 89$, $0.182 \times 512 = 93$ – an asymptote of **~93 beats in flight**. That is ONE channel's window, not eight: the harness's AXIS generator streams channels sequentially (finish one channel, then the next, so each channel's LFSR run stays contiguous for the golden CRC), so only one sink channel ever holds data, and the write engine's whole AW_MAX_OUTSTANDING = 8 window (8 x 8 = 64 beats, plus the bursts in their W phase) sits on that channel. Rapids ISSUE-006 reproduced this row in simulation (34.7 %, the board's number) and read it off the engine's outstanding counters. The source column reflects all eight channels reading concurrently, so the two columns are not comparable as measured; an interleaved-channel generator mode (rapids TASK-018) would put the sink on the same footing. STREAM's window on the same knobs is ~128 beats per channel (8 outstanding x 16-beat bursts) and its knee sits at 96-112 cycles. Wider bursts

remain a lever on both sides, and section 7.4 shows the DUT already runs 32- and 64-beat bursts at line rate.

Two notes from the instruments themselves. First, the latency columns are histogram means over log2 bins (bin b holds $[2^b, 2^{(b+1)})$), reported at its midpoint 1.5×2^b , and on this sweep every transaction of a row lands in one bin: the measured round trip is the injected delay plus roughly 24 cycles of DUT-plus-model overhead, so 0 lands in $[16,32)$, 8-32 in $[32,64)$, 48-96 in $[64,128)$, 128-192 in $[128,256)$, 256-384 in $[256,512)$ and 512 in $[512,1024)$ – the 24 / 48 / 96 / 192 / 384 / 768 staircase in the table is the histogram’s resolution, not a step in the DUT. Second, the v1.2 issue of this report carried `OBS_STICKY.HIST_SAMPLE_LOST` from 48 cycles up: the observer’s timestamp FIFO was sized `HIST_MAX_OUTSTANDING` = 8 per channel and the source read engine keeps more than that in flight once the delay exceeds one burst, so those v1.2 latency means came from a subset of transactions (and read low: 170, 264, 320, 530 where the full population sits at 192, 384, 384, 768). rapids TASK-012 raised the FIFO to 32 on the harness’s `u_obs_axi` (bitstream v4, post-route WNS +0.224 ns, still positive), the sweep was re-run, and `OBS_STICKY` reads 0 on all 12 rows for both observers. The utilization and bandwidth columns are identical between the two runs, which is what the design of the meters predicts: sample loss only ever touched the histograms.

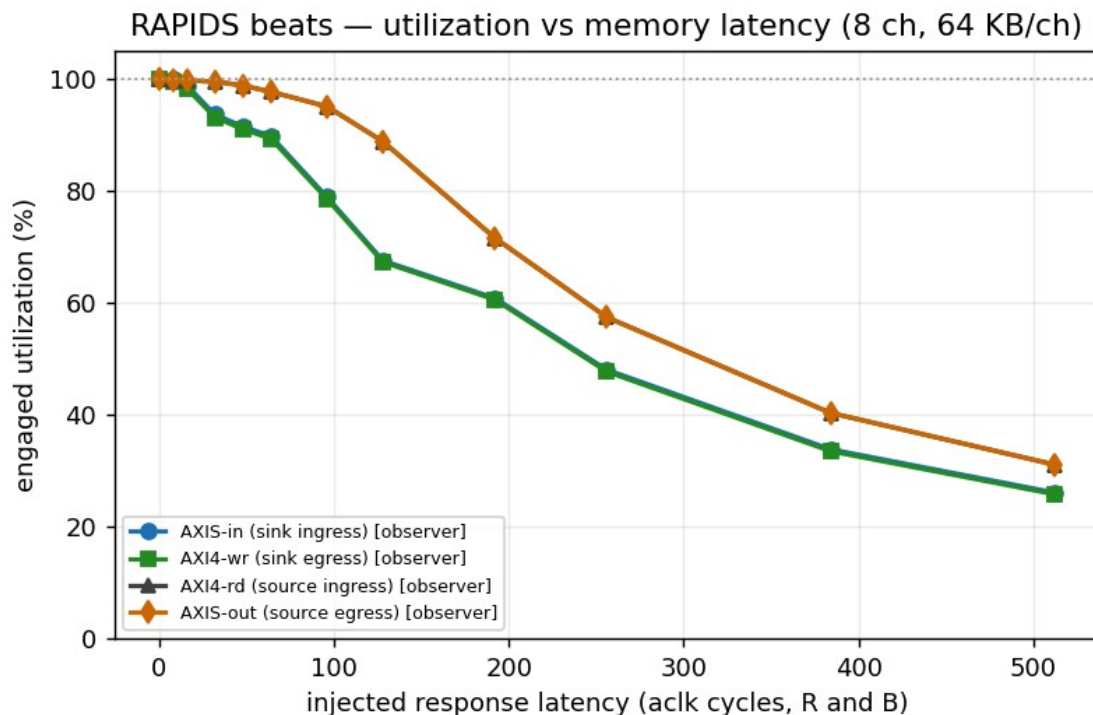


Figure 7.5 – utilization vs injected memory latency, all four observers.

7.6 What the observers add over the bare meters

Same utilization numbers (to the beat), plus what a meter cannot give: burst counts, the AR->first-R / AR->RLAST / AW->B latency distributions, exact AXIS bytes and packets per port with per-tid attribution, the tap’s own packet count as a cross-check, and a sticky bit that says when a number

undercounts. All of it through one register map shared with STREAM's harness, so the host code and the report generator are the same on both DMAs.

Appendix: data files & reproduce

File	Contents
perf/json/ genesys_obs_{A,B,C,E}.json	v1.2 observer campaign: descriptors x channels, burst length, size x channels, latency
perf/json/ genesys_full_matrix.json	channel × size matrix (v1.4, bare meters, PIPELINE = 1)
perf/json/ genesys_full_matrix_v1.1.json	the same matrix as measured for v1.1 (PIPELINE = 0 with the pre-BUG-005 engine)
perf/json/ genesys_8ch_*.json	earlier back-to-back runs (show the pre-fix wedge)
perf/plots/*.png	figures above (flows-rapids-beats/host/plot_char_reports.py)
<pre># full channel x size matrix in ONE programming (CHANNEL_RESET per run): source env_python python3 projects/fpga-systems/Genesys2/rapids_beats/flows-rapids-beats/host/run_characterization.py \ --port /dev/ttyUSB1 --channels 8 --suite \ --suite-channels 1,2,4,8 --suite-beats 4,16,64,256,1024,4096 -- suite-bp off \ --results projects/fpga-systems/Genesys2/rapids_beats/reports/perf/json/genesys_full_matrix.json</pre>	
<pre># figures: python3 projects/fpga-systems/Genesys2/rapids_beats/flows-rapids-beats/host/plot_char_reports.py \ --size projects/fpga-systems/Genesys2/rapids_beats/reports/perf/json/genesys_full_matrix.json \ --outdir projects/fpga-systems/Genesys2/rapids_beats/reports/perf/plots</pre>	
<pre># this report (DOCX + PDF, house style): cd projects/fpga-systems/Genesys2/rapids_beats/reports && ./generate_reports_pdf.sh --rev 1.0</pre>	

```

# v1.2 observer campaign (USE_OBSERVERS=1 bitstream: make bitstream
BOARD=genesys2 USE_OBSERVERS=1):
H=projects/fpga-systems/Genesys2/rapids_beats/flows-rapids-beats/host/
run_characterization.py
J=projects/fpga-systems/Genesys2/rapids_beats/reports/perf/json
python3 $H --port /dev/ttyUSB0 --channels 8 --suite --suite-bp off --
suite-seeds default \
    --suite-channels 1,2,4,8 --suite-beats 1024 --suite-descs
1,2,4,8,16 --results $J/genesys_obs_A.json
python3 $H --port /dev/ttyUSB0 --channels 8 --suite --suite-bp off --
suite-seeds default \
    --suite-channels 8 --suite-beats 1024 --suite-xfer
0,1,3,7,15,31,63 --results $J/genesys_obs_B.json
python3 $H --port /dev/ttyUSB0 --channels 8 --suite --suite-bp off --
suite-seeds default \
    --suite-channels 1,2,4,8 --suite-beats 1,4,16,64,256,1024,4096 --
results $J/genesys_obs_C.json
python3 $H --port /dev/ttyUSB0 --channels 8 --suite --suite-bp off --
suite-seeds default \
    --suite-channels 8 --suite-beats 1024 --suite-delay
0,8,16,32,48,64,96,128,192,256,384,512 --results $J/genesys_obs_E.json
# figures (size/channel plots from C; obs_desc_matrix /
obs_xfer_knee / obs_latency_knee from A / B / E):
python3 ../host/plot_char_reports.py --size $J/genesys_obs_C.json --
outdir ../reports/perf/plots
cd projects/fpga-systems/Genesys2/rapids_beats/reports &&
./generate_reports_pdf.sh --rev 1.3 --only perf

```

Genesys 2 host link: JTAG on the FT2232 (200300B818A0), UART on the separate FT232R (AU05X8RM, /dev/ttyUSB1); both must be connected at once, and the board must not be power-cycled between program and run.